

Convolution & Z-Transforms

A compact, from-zero IDSA revision guide

Second-year level • Black-and-white • 58 practice questions

What this covers. Discrete-time linear and circular convolution from your 14-page handwritten PDF, plus an introductory Z-transform extension. Learn the method, copy the worked example, then attempt the questions without looking. Draw on separate paper to keep this guide compact.

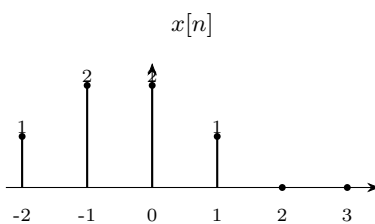
Source map. PDF pp. 1–4: indexed convolution; pp. 5–6: rectangular sequences; pp. 7–12 and 14: circular rotation; p. 13: comparison. The scans are out of teaching order. Z-transforms are added here, not present in the supplied notes. The guide targets these question families, not an unseen full syllabus.

1 Read a signal before calculating

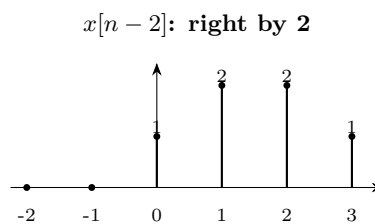
A discrete-time signal $x[n]$ is a list of values at integer indices n . The index is the position; $x[n]$ is the height. A **stem plot** puts a vertical line at each integer. Do not join sample tops as if time were continuous. Unless stated otherwise, finite lists are zero outside their listed indices.

$$\{1, 2, 2, 1\}_{n=-2,-1,0,1} \iff \begin{array}{c|cccc} n & -2 & -1 & 0 & 1 \\ \hline x[n] & 1 & 2 & 2 & 1 \end{array}$$

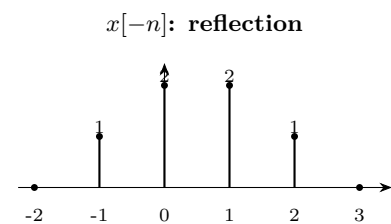
An arrow below an entry in the notes marks $n = 0$. Always convert it into an index row.



Horizontal labels: sample index.



Horizontal labels: sample index.



Horizontal labels: sample index.

The three building blocks

Impulse: $\delta[n] = 1$ at $n = 0$, and 0 elsewhere. Thus $\delta[n-r]$ is located at $n = r$.

Step: $u[n] = 1$ for $n \geq 0$, and 0 for $n < 0$. Here $u[0] = 1$.

Rectangle: $u[n] - u[n-L]$ equals 1 at $0, \dots, L-1$ and 0 elsewhere.

A safe way to move any sample

For $v[n] = x[an+b]$, an original sample at m moves to the solution of $an+b = m$, provided that solution is an integer. Its height stays the same. In this guide we mainly use $a = \pm 1$.

Try it now (answers on pages 11–12)

1. For the indexed $x[n]$ above, find $x[-1]$, $x[0]$, $x[3]$, and its nonzero index range.
2. List the nonzero values and indices of $x[n+1]$. Sketch them.
3. List and sketch $x[1-n]$. Use $1-n = m$.
4. List the nonzero samples of $\delta[n-2] + 2\delta[n+1]$.
5. List and sketch $u[n] - u[n-4]$.
6. Write the displayed $x[n]$ as a sum of shifted impulses.

2 Linear convolution: flip, shift, multiply, add

In a **linear time-invariant (LTI) system**, scaling/adding inputs scales/adds their outputs (linear), and delaying an input delays its output equally (time-invariant). Here $h[n]$ is the output caused by one unit impulse. Every input sample creates a scaled, shifted copy of $h[n]$; their sum is the output. This operation is convolution. The symbol $*$ does **not** mean ordinary pointwise multiplication.

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

n is the output position you are finding. k runs through the input positions. For finite lists, only overlapping nonzero samples contribute.

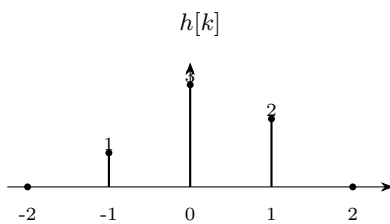
1. **Index and range.** If x occupies $a \dots b$ and h occupies $c \dots d$, calculate y on $a+c \dots b+d$. Its span has $L+M-1$ positions when the input spans have L and M positions.
2. **Flip** $h[k]$ into $h[-k]$. A sample at m moves to $-m$.
3. **Shift** to $h[n-k]$. That same sample now sits at $k = n-m$.
4. **Multiply** heights at matching k , then **add**. Repeat for each n .

Worked example A: exactly the indexed example in the notes

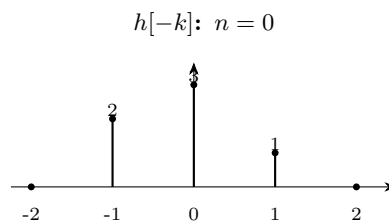
$$x = \{1, 2, 2, 1\}_{n=-2, -1, 0, 1}, \quad h = \{1, 3, 2\}_{n=-1, 0, 1}.$$

The output range is $[-2 + (-1), 1 + 1] = [-3, 2]$. A useful shortcut in *writing* the same sum is

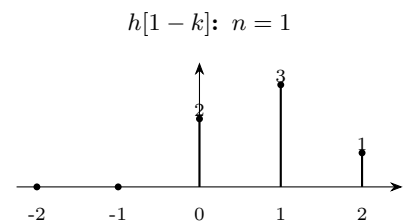
$$y[n] = x[-2]h[n+2] + x[-1]h[n+1] + x[0]h[n] + x[1]h[n-1].$$



Horizontal labels: sample index.



Horizontal labels: sample index.



Horizontal labels: sample index.

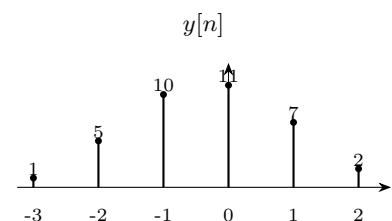
n	Nonzero overlap products	$y[n]$
-3	1(1)	1
-2	1(3) + 2(1)	5
-1	1(2) + 2(3) + 2(1)	10
0	2(2) + 2(3) + 1(1)	11
1	2(2) + 1(3)	7
2	1(2)	2

Answer: $y = \{1, 5, 10, 11, 7, 2\}_{n=-3, -2, -1, 0, 1, 2}$.

Check the total: $\sum y = (\sum x)(\sum h) = 6 \cdot 6 = 36$. The six outputs sum to 36. This catches many arithmetic errors, but cannot prove the order is correct.

Try it now (answers on pages 11–12)

7. Why is $y[-4] = y[3] = 0$ in this example?
8. Draw $h[-2-k]$ and use it to recalculate $y[-2]$.
9. If x occupies $-4 \dots 2$ and h occupies $1 \dots 3$, find the output range and span length.
10. For example A, find $(x * \delta[n-2])[n]$ and its range.



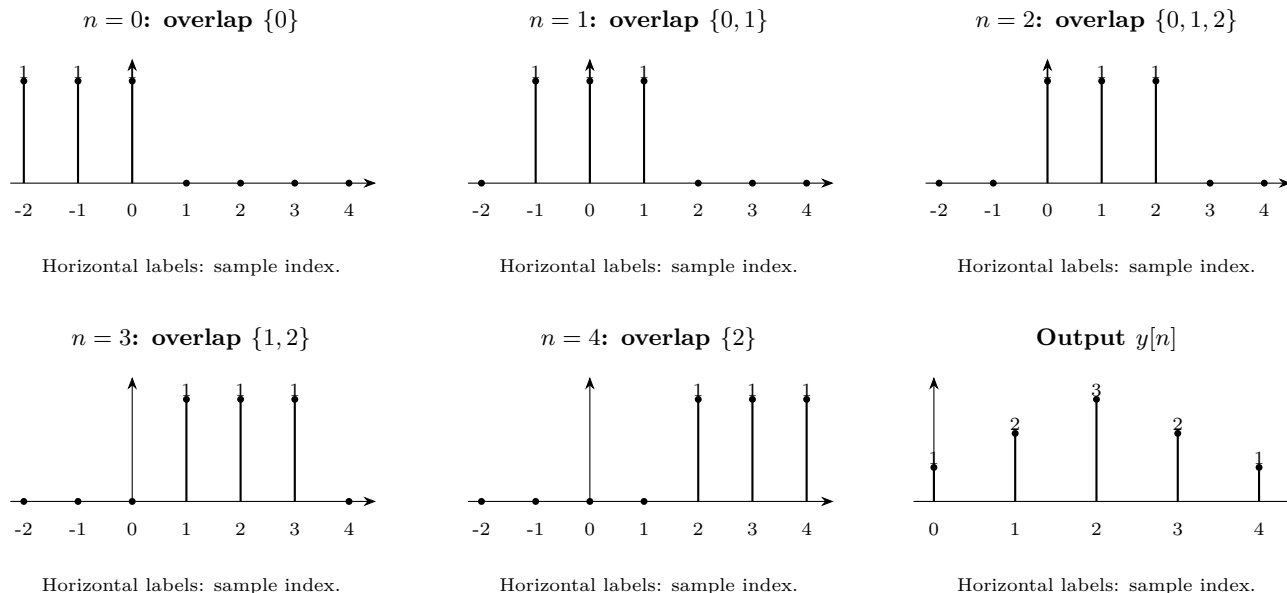
Horizontal labels: sample index.

3 See every overlap, then practise

Worked example B: the rectangle example from the notes

Let $x = h = \{1, 1, 1\}_{n=0,1,2}$. Both have length 3, so y has $3 + 3 - 1 = 5$ positions, $n = 0, \dots, 4$. Because every nonzero product is 1, $y[n]$ is just the **number of overlapping samples**.

In every panel below, keep $x[k] = 1$ at $k = 0, 1, 2$ fixed. The plotted stems show the moving $h[n - k]$. The bracketed set gives the overlapping k values.



$$y[0] = 1, \quad y[1] = 1 + 1 = 2, \quad y[2] = 1 + 1 + 1 = 3, \quad y[3] = 2, \quad y[4] = 1.$$

A second layout for the same multiply-add calculation

For $x = \{1, 2, 1\}$ and $h = \{1, -1\}$, both starting at zero, write one shifted row for each sample of x . Add down the columns. This is a check on the graphical method, not a different definition.

Output index	0	1	2	3
$x[0]h[n]$	1	-1	0	0
$x[1]h[n-1]$	0	2	-2	0
$x[2]h[n-2]$	0	0	1	-1
$y[n]$	1	1	-1	-1

Three fast checks and useful properties

Span: $L + M - 1$. **Ends:** first output = first-input product; last output = last-input product. **Sum:** $\sum y = (\sum x)(\sum h)$.

$$x * h = h * x, \quad x * (h_1 + h_2) = x * h_1 + x * h_2, \quad x * \delta[n - r] = x[n - r].$$

Avoid: forgetting the flip; dropping negative indices; shifting the heights rather than their positions; deleting interior zero samples; counting $L + M$ instead of $L + M - 1$.

Try it now (answers on pages 11–12)

All lists below start at $n = 0$ unless explicit indices are given. Sketch every output.

11. Find $\{1, 2\} * \{3, 4\}$. Show each overlap.
12. Find $\{1, 1, 1, 1\} * \{1, 1\}$ by counting overlaps.
13. Find $\{2, -1, 1\} * \{1, 2\}$. Check the sum.
14. Find $\{1, 2\}_{n=-1,0} * \{1, -1\}_{n=0,1}$. Mark the output origin.
15. Find $\{1, 0, 2\} * \{2, 1, 0\}$. Keep all five output positions.
16. Find $\{1, 2, 3\} * (\delta[n] + \delta[n - 2])$ using shifted copies.

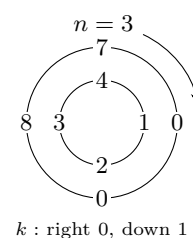
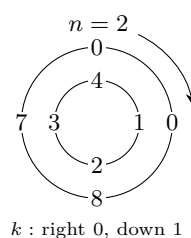
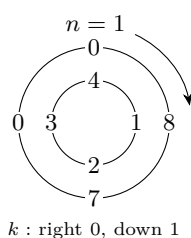
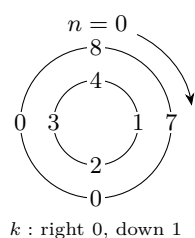
4 Circular convolution: the notes' rotating circles

An N -point circular convolution treats indices as positions on a ring: after $N - 1$, return to 0. For example, $(-1) \bmod 4 = 3$ and $5 \bmod 4 = 1$. The answer is one block of N values, repeated if a periodic extension is required.

$$y_N[n] = \sum_{k=0}^{N-1} x[k]h[(n-k) \bmod N], \quad 0 \leq n < N.$$

Worked example C: $x = \{1, 2, 3, 4\}$, $h = \{7, 8, 0, 0\}$, $N = 4$

1. Add zeros to obtain $h = \{7, 8, 0, 0\}$. Draw two concentric circles, divided into four positions ($360^\circ/4 = 90^\circ$ apart).
2. Place $x[0], x[1], x[2], x[3]$ **clockwise** on the inner circle, starting at the right. Keep it fixed. Place $h[0], h[1], h[2], h[3]$ **anticlockwise** on the outer circle, starting at the right. Opposite directions perform the flip.
3. Multiply aligned inner/outer values, then add: this is $y[0]$.
4. Rotate only the outer circle one position **clockwise** for $y[1]$. Repeat for $y[2]$ and $y[3]$.



Read each ring at fixed $k = 0, 1, 2, 3$ (right, down, left, up):

n	$h[n]$	$h[n-1]$	$h[n-2]$	$h[n-3]$	Dot product with $[1, 2, 3, 4]$
0	7	0	0	8	$1(7) + 2(0) + 3(0) + 4(8) = 39$
1	8	7	0	0	$1(8) + 2(7) + 3(0) + 4(0) = 22$
2	0	8	7	0	$1(0) + 2(8) + 3(7) + 4(0) = 37$
3	0	0	8	7	$1(0) + 2(0) + 3(8) + 4(7) = 52$

$$y_4 = \{39, 22, 37, 52\}$$

All h indices in this table are modulo 4. Every new row is a right rotation of the previous row.

Check: $39 + 22 + 37 + 52 = 150 =$
A one-line check for this exam

Try it now (answers on pages

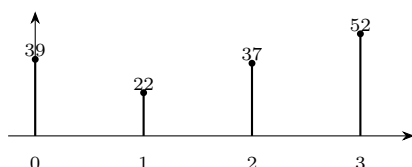
17. Evaluate $(-2) \bmod 4$, $(-5) \bmod 4$.

18. Write $h[(-k) \bmod 4]$ for $h = \{7, 8, 0, 0\}$.

19. Find the 4-point circular convolution of $x = \{1, 2, 3, 4\}$ and $h = \{7, 8, 0, 0\}$.

20. Find the 3-point circular convolution of $x = \{1, 2, 3\}$ and $h = \{7, 8, 0\}$.

$y_4[n]$: one period



Horizontal labels: sample index.

5 Linear versus circular: choose the right length

	Linear convolution	N -point circular convolution
Outside the block	Finite inputs are zero outside their ranges.	Indices wrap modulo N .
Output span	$L+M-1$ positions for listed input lengths.	Exactly N stored positions.
Graphical method	Flip, slide, multiply, add.	Reverse on a ring, rotate, multiply, add.
End behaviour	The tail stays at its own indices.	A tail past $N-1$ folds onto the start.
Transform link	Ordinary Z-transform multiplication.	N -point DFT multiplication; or polynomial multiplication with $z^{-N} = 1$.

Zero-padding rule: the essential correction to the notes

For finite lists, use the N stated in the question; add zeros to both lists up to that length (normally $N \geq \max(L, M)$). If no N is stated, a common convention is $N = \max(L, M)$; **state this assumption**. To make circular convolution equal linear convolution, choose

$$\boxed{N \geq L + M - 1} \quad \text{and zero-pad both inputs to length } N.$$

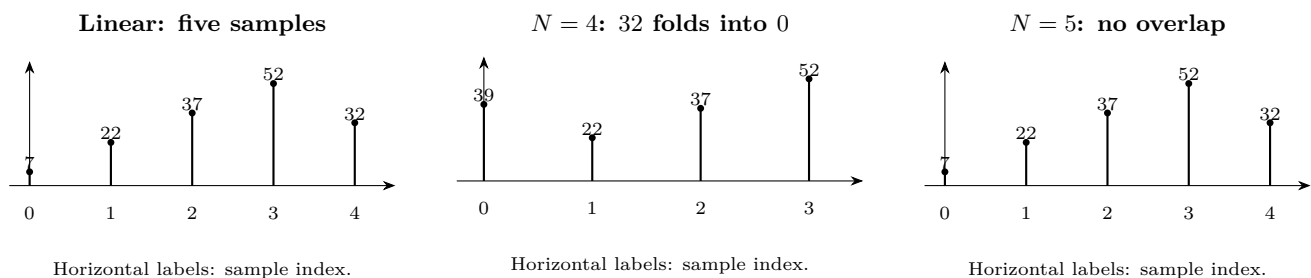
An LCM is useful when finding a common period of *already periodic* sequences. It is not a general zero-padding rule. Repeating $\{7, 8\}$ as $\{7, 8, 7, 8\}$ and padding it as $\{7, 8, 0, 0\}$ produce different answers. The numerical work in the notes uses the padded version.

Worked example D: watch the tail wrap

For the notes' $x = \{1, 2, 3, 4\}$ and $h = \{7, 8\}$, ordinary linear convolution gives

$$y_{\text{lin}} = \{7, 22, 37, 52, 32\} \quad (n = 0, \dots, 4).$$

At $N = 4$, index 4 wraps to index 0: $y_4[0] = 7 + 32 = 39$. The other samples stay put. At $N = 5$, nothing overlaps after wrapping, so the circular and linear outputs agree.



The middle plot shows one stored block, $n = 0, 1, 2, 3$. Its periodic extension has $y_4[4] = y_4[0] = 39$.

For zero-start finite inputs:
$$y_N[r] = \sum_{\ell \in \mathbb{Z}} y_{\text{lin}}[r + \ell N], \quad 0 \leq r < N.$$

Meaning: group linear-output indices by their remainder after division by N , and add each group. This overlap is called *time aliasing*. With sufficient padding it disappears. Both convolution types can be computed directly; FFT methods are not compulsory for circular convolution and can also compute padded linear convolution.

Try it now (answers on pages 11–12)

21. Input lengths are 5 and 4. Find the minimum N for circular convolution to reproduce the linear result.
22. Fold $\{1, 3, 5, 3\}$ into an $N = 3$ circular output.
23. Find both linear and 3-point circular convolutions of $\{1, 1, 1\}$ with itself.
24. Repeat example C with $h = \{7, 8, 7, 8\}$. Explain

why the answer changes.

25. For lengths 4 and 6, compare $\max(L, M)$, $\text{lcm}(L, M)$ and the minimum alias-free N .
26. Write five differences between linear and circular convolution. Include the condition under which their outputs agree.

6 Z-transform: attach a power to each sample

The Z-transform stores a sequence as an algebraic expression. The value becomes a coefficient; its index determines the power. z is a complex variable, but finite-list questions require only powers and arithmetic.

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Method: write the indices $\rightarrow$ write $x[n]z^{-n}$ for each nonzero sample $\rightarrow$ add $\rightarrow$ state where the expression converges (the ROC, explained next). Negative time gives a positive power of z .

Worked example E: transform the exact sequences from example A

n	-2	-1	0	1
$x[n]$	1	2	2	1
$x[n]z^{-n}$	z^2	$2z$	2	z^{-1}

$$X(z) = z^2 + 2z + 2 + z^{-1}, \quad H(z) = z + 3 + 2z^{-1}.$$

Their finite-plane ROC is $z \neq 0$. Finite sums cannot diverge except at singular points such as $z = 0$ when negative powers occur. Positive powers exclude the point at infinity if the extended complex plane is being used.

Why convolution becomes multiplication

Multiply one term from each transform: $x[k]z^{-k} h[m]z^{-m} = x[k]h[m]z^{-(k+m)}$. Terms with $k+m = n$ collect at z^{-n} . Their total coefficient is exactly $\sum_k x[k]h[n-k]$.

$$x * h \longleftrightarrow X(z)H(z)$$

For example A,

$$X(z)H(z) = z^3 + 5z^2 + 10z + 11 + 7z^{-1} + 2z^{-2}.$$

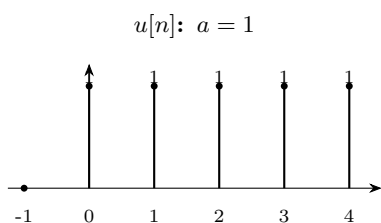
Read the indices $-3, -2, -1, 0, 1, 2$: the answer is again $\{1, 5, 10, 11, 7, 2\}$.

Worked example F: an infinite sequence using a geometric sum

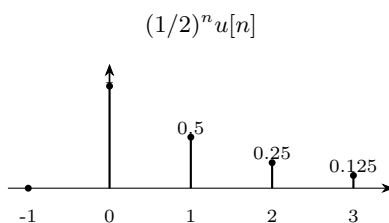
For $x[n] = a^n u[n]$ with $a \neq 0$, only $n \geq 0$ contributes:

$$X(z) = 1 + az^{-1} + a^2z^{-2} + \dots = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}, \quad |az^{-1}| < 1.$$

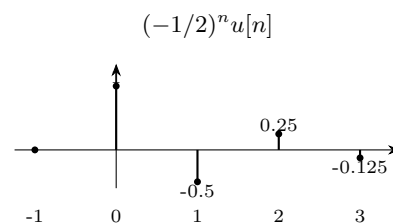
Thus $a^n u[n] \longleftrightarrow z/(z - a)$, $|z| > |a|$. The identity used is $1 + r + r^2 + \dots = 1/(1 - r)$ for $|r| < 1$; outside that condition, the series does not converge.



Horizontal labels: sample index.



Horizontal labels: sample index.



Horizontal labels: sample index.

Try it now (answers on pages 11–12)

27. Transform $\{2, -1, 3\}$, starting at $n = 0$.
28. Transform $\{2, -1, 3\}_{n=-1,0,1}$. Why does the answer change?
29. Transform $3\delta[n+2] - 2\delta[n-1]$.
30. Find the transform and ROC of $(1/3)^n u[n]$.
31. Find the transform and ROC of $(-2)^n u[n]$.
32. Multiply the transforms of $\{1, 2\}$ and $\{3, 4\}$, both starting at zero. Read off their convolution.

7 ROC, poles and the small transform table

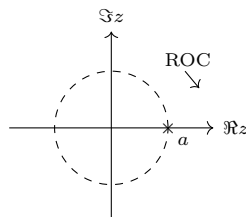
ROC = region of convergence: values of z for which the defining sum converges. A rational expression alone may represent different time sequences; its ROC tells you which one. A **pole** is a root of the denominator after cancellation; a **zero** is a root of the numerator after cancellation. The ROC never contains a pole.

Same expression, different sides of time ($a \neq 0$)

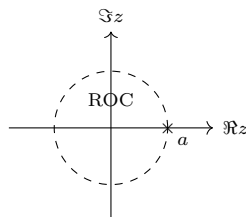
$$\frac{z}{z-a} \longleftrightarrow \begin{cases} a^n u[n], & |z| > |a| \quad (\text{right-sided}), \\ -a^n u[-n-1], & |z| < |a| \quad (\text{left-sided}). \end{cases}$$

For the left-sided case, write $n = -m$, $m \geq 1$: $X(z) = -\sum_{m=1}^{\infty} (z/a)^m = z/(z-a)$ when $|z/a| < 1$. The minus sign and the endpoint $n = -1$ matter.

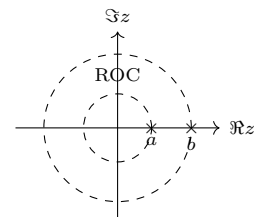
Right-sided: $|z| > a$



Left-sided: $|z| < a$



Two-sided: $a < |z| < b$



These schematic plots use positive real poles $0 < a < b$. Crosses mark poles; dashed boundary circles are excluded. An ROC is a disk, exterior, or annulus centred at the origin, not just a region near a real-axis pole.

$x[n]$	$X(z)$	ROC (finite z)
$\delta[n]$	1	All z
$\delta[n-r]$, $r > 0$	z^{-r}	$z \neq 0$
$u[n]$	$1/(1-z^{-1})$	$ z > 1$
$a^n u[n]$, $a \neq 0$	$1/(1-az^{-1})$	$ z > a $
$-a^n u[-n-1]$, $a \neq 0$	$1/(1-az^{-1})$	$ z < a $
$na^n u[n]$, $a \neq 0$	$az^{-1}/(1-az^{-1})^2$	$ z > a $
$(n+1)a^n u[n]$, $a \neq 0$	$1/(1-az^{-1})^2$	$ z > a $

System interpretation. A causal impulse response has $h[n] = 0$ for $n < 0$. An LTI system is BIBO stable if $\sum_n |h[n]| < \infty$, equivalently its ROC contains the unit circle $|z| = 1$. For a causal rational system, this requires every uncanceled pole to lie strictly inside the unit circle. A growing sequence can still have a Z-transform.

Try it now (answers on pages 11–12)

33. Invert $z/(z-1/2)$ for $|z| > 1/2$ and for $|z| < 1/2$.

34. Find the poles, zero and ROC of the causal $H(z) = z/(z-0.8)$. Is the system stable?

35. Give all possible ROCs of $z^2/[(z-1/2)(z-2)]$. Which contain the unit circle?

36. If $h[n] = 2^n u[n]$, is the system causal? Stable? Does $H(z)$ exist?

37. What is the finite-plane ROC of $X(z) = 1 + 2z^{-1}$? Of $X(z) = z^2$?

38. Find $\mathcal{Z}\{n(1/2)^n u[n]\}$ and its ROC.

8 Inverse Z-transform: choose the shortest method

Decision: finite powers $\rightarrow$ read coefficients; one standard fraction $\rightarrow$ table; product of simple factors $\rightarrow$ partial fractions; a few initial samples $\rightarrow$ power series. Always read the ROC before selecting right- or left-sided sequences.

Method 1: read coefficients, with indices

$$X(z) = 2z + 3 - z^{-1} + 4z^{-3} \Rightarrow x = \{2, 3, -1, 0, 4\}_{n=-1,0,1,2,3}.$$

The missing z^{-2} term means $x[2] = 0$. Equivalently, $x[n] = 2\delta[n+1] + 3\delta[n] - \delta[n-1] + 4\delta[n-3]$.

Method 2: partial fractions in z^{-1}

Find $x[n]$ for

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})}, \quad |z| > \frac{1}{2}.$$

Write $q = z^{-1}$ temporarily. Split into the exact table shapes:

$$\frac{1}{(1 - \frac{1}{2}q)(1 - \frac{1}{4}q)} = \frac{A}{1 - \frac{1}{2}q} + \frac{B}{1 - \frac{1}{4}q}.$$

Multiply by both denominators: $1 = A(1 - \frac{1}{4}q) + B(1 - \frac{1}{2}q)$. Comparing the constant and q coefficients gives $A+B = 1$ and $A/4 + B/2 = 0$. Thus $A = 2$, $B = -1$. Both terms are right-sided because the ROC is outside both poles:

$$x[n] = [2(1/2)^n - (1/4)^n]u[n].$$

Check: $x[0] = 2 - 1 = 1$. The first samples are 1, 3/4, 7/16, 15/64. If the ROC were $1/4 < |z| < 1/2$, the first term would be left-sided:

$$x[n] = -2(1/2)^n u[-n-1] - (1/4)^n u[n].$$

Method 3: long division / matching powers

For a causal sequence with $X(z) = (1 + z^{-1})/(1 - \frac{1}{2}z^{-1})$, let $q = z^{-1}$ and set $X = c_0 + c_1q + c_2q^2 + \dots$. Then

$$(1 - \frac{1}{2}q)(c_0 + c_1q + c_2q^2 + \dots) = 1 + q.$$

Match equal powers: $c_0 = 1$, $c_1 - \frac{1}{2}c_0 = 1$, and $c_n - \frac{1}{2}c_{n-1} = 0$ for $n \geq 2$. So $x[0], x[1], x[2], x[3] = 1, 3/2, 3/4, 3/8$. This is ordinary algebra, with the sample index stored in the power.

Repeated pole: do not force two identical simple fractions

$$\frac{1}{(1 - az^{-1})^2} \longleftrightarrow (n+1)a^n u[n] \quad (|z| > |a|).$$

You can see the factor $n+1$ by convolving $a^n u[n]$ with itself: there are $n+1$ equal products a^n for $n \geq 0$. A general repeated factor uses both $A/(1 - aq)$ and $B/(1 - aq)^2$.

Try it now (answers on pages 11–12)

39. Invert $2 + 3z^{-1} - z^{-3}$. Include the missing sample.

40. Invert $z^2 + 2z + 3$. Give the index range.

41. Invert $z^{-2}/(1 - \frac{1}{2}z^{-1})$, ROC $|z| > 1/2$.

42. Invert $1/[(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})]$, ROC $|z| > 1/2$.

Show A, B .

43. Invert $1/(1 - \frac{1}{2}z^{-1})^2$ for $|z| > 1/2$. Give the first three samples.

44. Use matching powers to find the first four causal samples of $(1 - z^{-1})/(1 - \frac{1}{2}z^{-1})$.

9 Use Z-transforms to solve system questions

Operation	Z-domain rule	What to watch
$\alpha x[n] + \beta v[n]$	$\alpha X(z) + \beta V(z)$	Use a common ROC; cancellation may enlarge it.
$x[n - r]$	$z^{-r} X(z)$	Delay moves the start to r .
$x[-n]$	$X(z^{-1})$	ROC radii invert; endpoints need care.
$a^n x[n], a \neq 0$	$X(z/a)$	ROC radii scale by $ a $.
$nx[n]$	$-z dX/dz$	Differentiate carefully.
$x * h$	$X(z)H(z)$	ROC includes the common convergence region.

Shift and convolution rules here use the **bilateral** transform (sum over all integers). A finite shift can change inclusion of 0 or infinity. Check the final expression and actual sequence.

Worked example G: delay without an exponent error

$$\mathcal{Z}\{(1/2)^{n-2}u[n-2]\} = \frac{z^{-2}}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}.$$

The first sample is 1 at $n = 2$. If the question instead gives $(1/2)^n u[n-2]$, factor out $(1/2)^2 = 1/4$ first; its transform is *one quarter* of the displayed expression.

Worked example H: difference equation, initially at rest

A difference equation relates present output to past output and input. Given

$$y[n] - \frac{1}{2}y[n-1] = x[n], \quad x[n] = u[n], \quad y[n] = 0 \ (n < 0),$$

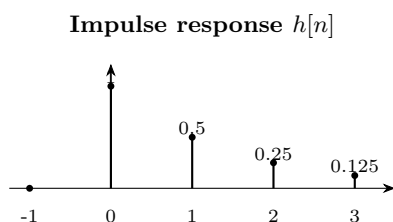
take transforms, collect the Y terms, and divide:

$$Y - \frac{1}{2}z^{-1}Y = X, \quad H = \frac{Y}{X} = \frac{1}{1 - \frac{1}{2}z^{-1}}, \quad Y = \frac{1}{(1 - z^{-1})(1 - \frac{1}{2}z^{-1})}.$$

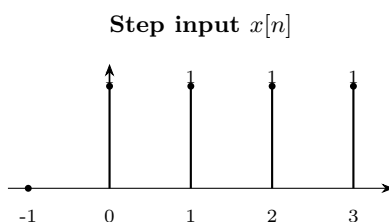
Partial fractions give $Y = 2/(1 - z^{-1}) - 1/(1 - \frac{1}{2}z^{-1})$. Causality selects $|z| > 1$, hence

$$y[n] = (2 - (1/2)^n)u[n].$$

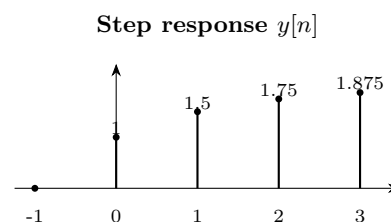
Check in time: $y[0] = 1$, $y[1] = 1 + \frac{1}{2}(1) = 1.5$, $y[2] = 1 + \frac{1}{2}(1.5) = 1.75$.



Horizontal labels: sample index.



Horizontal labels: sample index.



Horizontal labels: sample index.

If an initial value is given: use the one-sided rule

The unilateral transform is $Y_+(z) = \sum_{n=0}^{\infty} y[n]z^{-n}$. For equations valid at $n \geq 0$,

$$\mathcal{Z}_+\{y[n-1]\} = z^{-1}Y_+(z) + y[-1].$$

Thus $y[n] - ay[n-1] = x[n]$ gives $Y_+ = (X_+ + ay[-1])/(1 - az^{-1})$. Use $H = Y/X$ as a system transfer function only for the zero-initial-condition response.

Try it now (answers on pages 11–12)

45. Find the transform of $(1/2)^n u[n-2]$ and its ROC.
 46. Find $\{1, 1\} * \{1, 2, 1\}$ by multiplying their transforms. Both start at zero.
 47. Initially at rest, solve $y[n] - \frac{1}{3}y[n-1] = \delta[n]$.

48. Initially at rest, solve $y[n] - \frac{1}{2}y[n-1] = u[n]$. Verify the first three samples independently.
 49. For $n \geq 0$, solve $y[n] - \frac{1}{2}y[n-1] = 0$ with $y[-1] = 2$.
 50. Invert $H(z) = 1/(1 - 2z^{-1})$ with ROC $|z| < 2$. Is this impulse response causal? Stable?

10 Mixed exam practice and a last-minute checklist

Suggested use: allow 50–60 minutes for Q51–58. Show indices, the method, a boxed result and a check. For graphs, label both axes and the origin. Treat the listed marks as practice weights, not your university’s marking scheme.

51. Linear convolution [6]. Let $x = \{1, 2, 1\}_{n=-1,0,1}$ and $h = \{1, -1\}_{n=0,1}$. Find $y = x * h$ by flip–shift–multiply–add. Draw x , $h[-k]$, and y . Check the output sum.

52. Circular convolution [6]. Find the 4-point circular convolution of $x = \{2, 0, 1, 1\}$ and $h = \{1, 2, 0, 0\}$. Show all four rotated rows.

53. Padding and wrapping [5]. Find the linear convolution of the unpadded lists $\{2, 0, 1, 1\}$ and $\{1, 2\}$. Use it to check Q52. State the minimum padding length that reproduces the linear output.

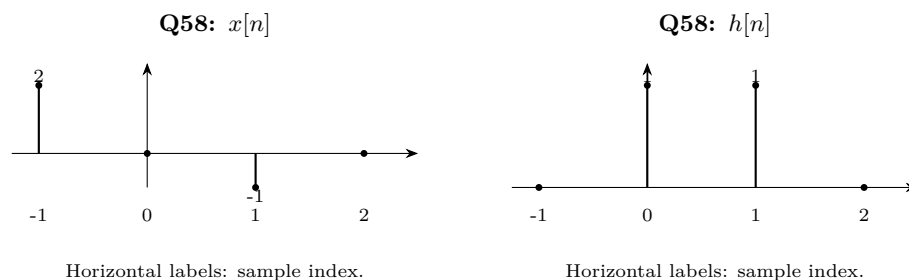
54. Transform an indexed list [4]. Find $X(z)$ and its finite-plane ROC for $x = \{1, -2, 0, 3\}_{n=-1,0,1,2}$. Recover the sequence by reading powers.

55. Inverse and ROC [6]. Invert $X(z) = 1/[(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})]$ for (a) $|z| > 2$ and (b) $1/2 < |z| < 2$. If each result is an impulse response, identify causality and stability.

56. System response [6]. Initially at rest, $y[n] - \frac{1}{4}y[n-1] = x[n]$. Find $H(z)$, $h[n]$, and the response to $x[n] = u[n]$. Verify $y[0]$ and $y[1]$.

57. Find the mistakes [4]. A student says: “Convolution means multiplying matching samples; the output length is $L + M$; the LCM always determines zero-padding; every Z-transform fraction uniquely determines its sequence without an ROC.” Correct each statement.

58. Graph-to-answer [5]. The two plots below define finite sequences (zero elsewhere). Read their index tables, find their linear convolution, then find its Z-transform.



Write these from memory before the exam

Trigger in the question	Your next move
“Convolve” finite signals	Index $\rightarrow$ range $\rightarrow$ flip $h \rightarrow$ shift $\rightarrow$ multiply overlaps $\rightarrow$ add.
“ N -point circular”	Pad to stated $N \rightarrow$ opposite circle directions $\rightarrow$ rotate outer circle clockwise $\rightarrow$ dot products.
“Linear using circular”	Pad both to $N \geq L + M - 1$.
“Find $X(z)$ ”	Attach z^{-n} to each sample; use geometric series for exponentials; give ROC.
“Find inverse”	Read finite powers or split into table fractions; select sides using ROC.
“System / recurrence”	State initial conditions; collect Y ; solve algebraically; invert; verify initial samples.

Final checks: origin marked; negative-index powers correct; output span correct; $\sum y = (\sum x)(\sum h)$; circular length stated; poles outside ROC; delayed sequences start at the correct sample. Stability needs the unit circle in the ROC, not merely an expression that looks finite.

11 Answer key: foundations and convolution

Notation in this key: unlabelled lists begin at $n = 0$. Each entry gives its value at one successive integer. Graphs follow directly from the listed index/value pairs. Try the questions before reading the answers.

1. $2, 2, 0$; nonzero range $-2 \leq n \leq 1$.
2. $\{1, 2, 2, 1\}_{n=-3, -2, -1, 0}$. An advance by 1 moves every sample left by 1.
3. $\{1, 2, 2, 1\}_{n=0, 1, 2, 3}$. Solve $n = 1 - m$; put the answers in increasing n .
4. $x[-1] = 2, x[2] = 1$; all others zero.
5. 1 at $n = 0, 1, 2, 3$, zero elsewhere. The sample at $n = 4$ is $1 - 1 = 0$.
6. $\delta[n + 2] + 2\delta[n + 1] + 2\delta[n] + \delta[n - 1]$.
7. There is no overlap outside the output range $-3 \dots 2$.
8. $h[-2 - k]$ has heights $2, 3, 1$ at $k = -3, -2, -1$. Only $k = -2, -1$ overlap: $1(3) + 2(1) = 5$.
9. Range $-3 \dots 5$; length $7 + 3 - 1 = 9$.
10. $x[n - 2] = \{1, 2, 2, 1\}_{n=0, 1, 2, 3}$.
11. $\{3, 10, 8\}$: $1(3), 1(4) + 2(3), 2(4)$. Sum $21 = 3 \cdot 7$.
12. $\{1, 2, 2, 2, 1\}$. The two-sample window has two overlaps at the middle three positions.
13. $\{2, 3, -1, 2\}$. Middle terms: $2(2) - 1 = 3$ and $(-1)(2) + 1 = -1$. Sum $6 = (2)(3)$.
14. $\{1, 1, -2\}_{n=-1, 0, 1}$. The middle sample is at the origin.
15. $\{2, 1, 4, 2, 0\}$. Five listed positions; the actual last nonzero sample is at $n = 3$ because the supplied h has a trailing zero.
16. $x[n] + x[n - 2] = \{1, 2, 4, 2, 3\}$.
17. $2, 3, 0$. Add or subtract multiples of 4 until the result lies in $\{0, 1, 2, 3\}$.
18. $\{1, 4, 3, 2\}$. Circular reversal keeps the zero-index entry first.
19. $\{5, 3, 5, 7\}$. Rotated h rows are $(1, 0, 0, 1), (1, 1, 0, 0), (0, 1, 1, 0), (0, 0, 1, 1)$.
20. $\{1, 1, -2\}$. Rows $(1, 0, -1), (-1, 1, 0), (0, -1, 1)$ dotted with $(1, 2, 0)$.
21. $N_{\min} = 5 + 4 - 1 = 8$.
22. $\{4, 3, 5\}$. Index 3 folds onto index 0, giving $1 + 3 = 4$.
23. Linear: $\{1, 2, 3, 2, 1\}$. Circular: $\{3, 3, 3\}$, because $1 + 2 = 3$ and $2 + 1 = 3$ at indices 0 and 1.
24. $\{76, 74, 76, 74\}$. For instance $y[0] = 1(7) + 2(8) + 3(7) + 4(8) = 76$. Repetition inserts extra nonzero samples; it is not zero-padding.
25. $\max = 6$, $\text{LCM} = 12$, minimum alias-free length $= 4 + 6 - 1 = 9$. Six allows wrapping; nine suffices; twelve works but is unnecessary.
26. Use the five rows on page 5. Equality with linear convolution requires zero-padding to at least $L + M - 1$ (with extra output zeros if N is larger). Circular convolution has wraparound by definition; aliasing relative to the linear answer occurs only when nonzero contributions fold together.

Why the source's circular example needs care

The supplied notes mention an LCM and also display repeated values, but the worked products use $h = \{7, 8, 0, 0\}$. This guide preserves that worked method and the verified answer $\{39, 22, 37, 52\}$. Q24 shows the different answer obtained if the repeated values were actually used.

One more distinction: $L + M - 1$ is the full output array length for finite input arrays of lengths L, M . End zeros in an input or cancellations can reduce the range of *nonzero* output samples. Retain requested array positions when presenting an exam answer.

12 Answer key: Z-transforms and mixed exam

27. $X = 2 - z^{-1} + 3z^{-2}$; ROC $z \neq 0$.
28. $X = 2z - 1 + 3z^{-1}$; ROC $z \neq 0$. Moving the origin changes the powers.
29. $X = 3z^2 - 2z^{-1}$; ROC $z \neq 0$.
30. $X = 1/(1 - \frac{1}{3}z^{-1})$, $|z| > 1/3$.
31. $X = 1/(1 + 2z^{-1})$, $|z| > 2$. The pole is at -2 , but its radius is 2.
32. $(1+2z^{-1})(3+4z^{-1}) = 3+10z^{-1}+8z^{-2}$; output $\{3, 10, 8\}$.
33. Outside: $(1/2)^n u[n]$. Inside: $-(1/2)^n u[-n-1]$.
34. Pole 0.8, zero 0, ROC $|z| > 0.8$. Stable: the unit circle is in the ROC.
35. $|z| > 2$, $1/2 < |z| < 2$, or $|z| < 1/2$. Only the middle ROC contains $|z| = 1$.
36. Causal, unstable. The transform exists for $|z| > 2$; its ROC excludes the unit circle.
37. $1 + 2z^{-1}$: $z \neq 0$. z^2 : all finite z , including 0. The latter excludes infinity in the extended plane.
38. $X = (\frac{1}{2}z^{-1})/(1 - \frac{1}{2}z^{-1})^2$, $|z| > 1/2$.
39. $\{2, 3, 0, -1\}$, at $n = 0, 1, 2, 3$.
40. $\{1, 2, 3\}_{n=-2, -1, 0}$.
41. $(1/2)^{n-2} u[n-2]$; first nonzero sample 1 at $n = 2$.
42. $A + B = 1$, $A/3 + B/2 = 0$, so $A = 3$, $B = -2$. $x[n] = [3(1/2)^n - 2(1/3)^n] u[n]$.
43. $(n+1)(1/2)^n u[n]$; first samples 1, 1, 3/4.
44. 1, $-1/2$, $-1/4$, $-1/8$. Coefficients obey $c_0 = 1$, $c_1 - \frac{1}{2}c_0 = -1$, $c_n = \frac{1}{2}c_{n-1}$ for $n \geq 2$.
45. $X = (\frac{1}{4}z^{-2})/(1 - \frac{1}{2}z^{-1})$, $|z| > 1/2$.
46. $Y = (1 + z^{-1})(1 + 2z^{-1} + z^{-2}) = 1 + 3z^{-1} + 3z^{-2} + z^{-3}$; $y = \{1, 3, 3, 1\}$.
47. $Y = 1/(1 - \frac{1}{3}z^{-1})$, so $y[n] = (1/3)^n u[n]$, ROC $|z| > 1/3$.
48. $y[n] = [2 - (1/2)^n] u[n]$. Direct recursion gives 1, 3/2, 7/4.

ROC $|z| > 1$.

49. $Y_+ = 1/(1 - \frac{1}{2}z^{-1})$; $y[n] = (1/2)^n$ for $n \geq 0$. Indeed $y[0] = \frac{1}{2}(2) = 1$. The specified prehistory is $y[-1] = 2$.

50. $h[n] = -2^n u[-n-1]$. Noncausal, stable: $\sum_{n=-\infty}^{-1} 2^n = 1$, and $|z| = 1$ lies inside $|z| < 2$.

51. $y = \{1, 1, -1, -1\}_{n=-1, 0, 1, 2}$. Products: 1; $-1+2$; $-2+1$; -1 . Sum $0 = (4)(0)$. $h[-k]$ has -1 at $k = -1$ and 1 at $k = 0$.

52. Rows $(1, 0, 0, 2)$, $(2, 1, 0, 0)$, $(0, 2, 1, 0)$, $(0, 0, 2, 1)$ dotted with $(2, 0, 1, 1)$ give $\{4, 4, 1, 3\}$.

53. Linear output $\{2, 4, 1, 3, 2\}$. Wrap the last 2 onto the first 2 to get Q52. Minimum $N = 4 + 2 - 1 = 5$.

54. $X = z - 2 + 3z^{-2}$, ROC $z \neq 0$. Coefficients at $n = -1, 0, 1, 2$ are 1, $-2, 0, 3$.

55. Let $q = z^{-1}$:

$$X = \frac{-1/3}{1 - \frac{1}{2}q} + \frac{4/3}{1 - 2q}.$$

(a) $x[n] = [-\frac{1}{3}(1/2)^n + \frac{4}{3}2^n] u[n]$: causal, unstable.

(b) $x[n] = -\frac{1}{3}(1/2)^n u[n] - \frac{4}{3}2^n u[-n-1]$: noncausal, stable (ROC contains the unit circle).

56. $H = 1/(1 - \frac{1}{4}z^{-1})$, $h[n] = (1/4)^n u[n]$, ROC $|z| > 1/4$.

$$Y = \frac{4/3}{1 - z^{-1}} - \frac{1/3}{1 - \frac{1}{4}z^{-1}}.$$

$y[n] = [\frac{4}{3} - \frac{1}{3}(1/4)^n] u[n]$, ROC $|z| > 1$. Checks: $y[0] = 1$, $y[1] = 5/4$.

57. (i) Convolution sums products after reversal and shifting. (ii) Linear array length is $L + M - 1$. (iii) Use stated N ; for no aliasing use $N \geq L + M - 1$. (iv) A rational bilateral Z-transform needs an ROC or equivalent sidedness information.

58. $x = \{2, 0, -1\}_{n=-1, 0, 1}$, $h = \{1, 1\}_{n=0, 1}$. Output $y = \{2, 2, -1, -1\}_{n=-1, 0, 1, 2}$. $X = 2z - z^{-1}$, $H = 1 + z^{-1}$, so $Y = 2z + 2 - z^{-1} - z^{-2}$, ROC $z \neq 0$.